

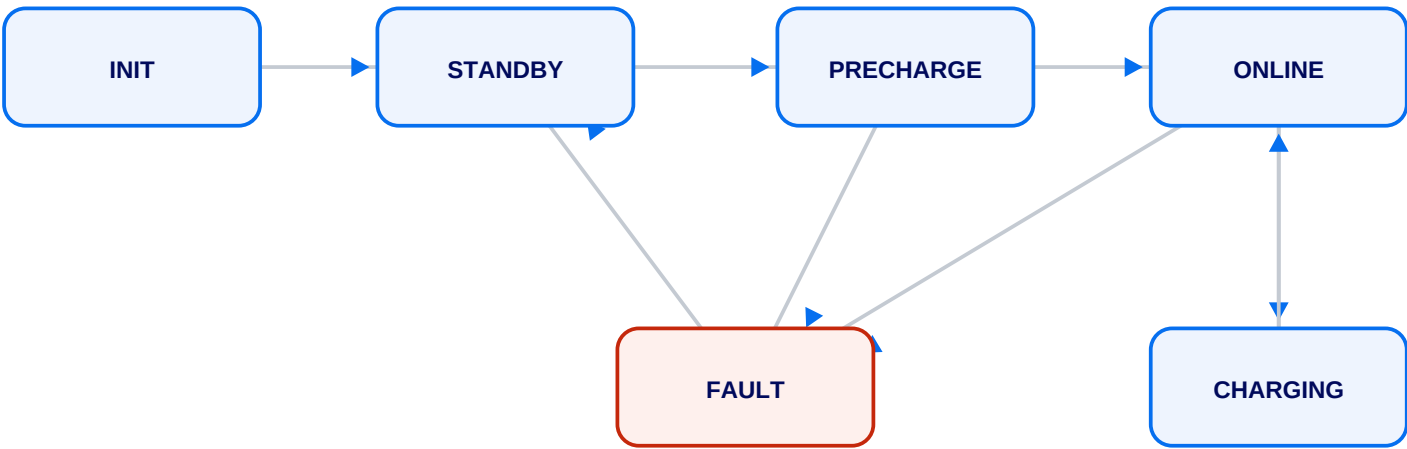
High-Voltage Battery Management System Specification

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This specification defines the contactor sequencing, diagnostic coverage, communication signals, and calibration parameters for the High-Voltage Battery Management System (BMS). The contactor state machine governs transitions between standby, precharge, online, charging, and fault handling. All diagnostic trouble codes are debounced per the table below.

1. Contactor State Machine

The module sequences the HV contactors through the following states. Note the precharge timeout path and the multiple paths into FAULT — reconstructing this from prose alone is error-prone.



2. Diagnostic Trouble Codes (DTC Matrix)

DTC Code	Description	Fault Action	MIL	Debounce (ms)
P0A80	Replace Hybrid Battery Pack	Limit power, warn	Yes	1000
P0AFA	Battery Pack Voltage Low	Open contactors	Yes	200
P0A0D	HV Interlock Circuit Open	Inhibit contactor close	Yes	50
P0AC0	Cell Over-Temperature	Force cooling, derate	No	1500
P0A94	DC/DC Converter Performance	Derate aux loads	No	800
P0A1F	Battery Energy Control Module Fault	Safe-state, latch	Yes	100
U029E	Lost Comms with Cell Sense Module	Use last known values	Yes	400

3. CAN Signal Definitions

Signal Name	Message ID	Start Bit	Length	Scale	Unit	Cycle (ms)
BMS_PackVoltage	0x3C0	0	16	0.1	V	10
BMS_PackCurrent	0x3C0	16	16	0.05	A	10
BMS_StateOfCharge	0x3C2	0	10	0.1	%	100
BMS_MaxCellTemp	0x3C2	16	8	1	degC	100
BMS_MinCellVoltage	0x3C4	0	12	0.001	V	50
BMS_ContactorState	0x3C8	0	4	1	enum	100
BMS_IsolationResistance	0x3C8	8	12	1	kOhm	1000

4. Calibration Parameters

Parameter ID	Name	Min	Max	Default	Unit
K_SOC_RESERVE	Usable SOC Reserve	2	15	5	%
V_CELL_MAX	Max Cell Voltage	4.0	4.25	4.2	V
V_CELL_MIN	Min Cell Voltage	2.5	3.2	3.0	V
T_CHG_MAX	Max Charge Temperature	40	60	50	degC
I_CHG_MAX	Max Charge Current	50	250	200	A
T_PRECHG_TO	Precharge Timeout	100	800	400	ms